

Year 9

Mini Test 1

Number

Name: Marking key.

Total Marks: 30 marks

Total Time: 25 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

1. Choose a number from the box that is:

[5 marks]

a. A prime number

7

b. A square number

9 or 1

c. A rational number

all except $\sqrt{2}$.

d. An irrational number

$\sqrt{2}$

e. A factor of 15

1

$\frac{1}{3}$	$\sqrt{16}$	9	1	$\sqrt{2}$
0.45	-5	$\frac{2}{5}$	7	0.4

2. Write the correct spelling for the following misspelt words.

[4 marks]

Multiple

Multiple.

Qube root

Cube root

Inproper fraction

Improper Frachon.

Recurring decimal

Recurring decimal!

3. Evaluate the following

[5 Marks]

a. $-7 + (-3) = \underline{-10}$

b. $(-20) \div (-4) = \underline{5}$

c. $-2 \times (6 - 9) = \underline{6}$

d. $3 + 4 \times (-2) - 5 = \underline{-10.}$

e. $6^2 - 3 \times (-2) = \underline{42.}$

4. Which of the following is 7.93642 rounded to 2 decimal places (circle the correct answer)

[1 mark]

a. 7.93

b. 7.94

c. 7.930

d. 7.94000

5. Which of the following is 12.978 rounded to 1 decimal places (circle the correct answer)

[1 mark]

a. 12.9

b. 13

c. 13.0

d. 12.10

6. Round the following numbers to 3 significant figures

[3 marks]

a. 2 934 065 $2930\ 000$

b. 67 054 67100

c. 0.007 639 42 $0.00764.$ 2

7. Find

[2 marks]

a. $\frac{3}{5} + \frac{4}{7} = \frac{21}{35} + \frac{20}{35} = \frac{41}{35} = 1\frac{6}{35}$

b. $\frac{2}{3} \times 1\frac{5}{6} = \frac{2}{3} \times \frac{11}{6} = \frac{11}{9} = 1\frac{2}{9}$

8. Simplify the following ratios

[3 marks]

a. $5 : 20$ $1 : 4$

b. $\frac{1}{4} : \frac{2}{3}$ $3 : 8$

c. $15\text{mins} : 1\text{ hour}$

$1 : 4$

9. \$42 is divided between James and Mark in a ratio of 2 : 5.

[2 marks]

a. How much does James get? $\$12.$

b. How much more does Mark get than James? $\$18.$

10. Write these rates in simplest form:

[2 marks]

a. 42 m in 6 seconds

7m/s

b. 5000 Litres in $2\frac{1}{2}$ hours

2000L/h

11. A car travels at an average speed of 80km/h. How far does the car travel in:

[2 marks]

a. $2\frac{1}{2}$ hours?

200km.

b. 40 minutes?

$53\frac{1}{3}$ km